

Acoustic Analysis – Room 238 Mashburn Hall

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Caleb Milton Young
*Department of Physics,
Astronomy, and Engineering
University of Central Arkansas*
Conway, United States of
America
cyoung24@cub.uca.edu

Eric Dean LeJune
*Department of Physics,
Astronomy, and Engineering
University of Central Arkansas*
Conway, United States of
America
elejune@cub.uca.edu

Edward Jaleel Fields
*Department of Physics,
Astronomy, and Engineering
University of Central Arkansas*
Conway, United States of
America
efields@cub.uca.edu

Brandon Donaldson
*Department of Physics,
Astronomy, and Engineering
University of Central Arkansas*
Conway, United States of
America
bdonaldson1@cub.uca.edu

Abstract: This paper covers the final project done in Dr. Slaton's Spring 2026 Acoustics class. The experiment, analysis, and results involved in this project will be discussed below. The goal is to provide an acoustic analysis of room 238 in Mashburn Hall at UCA.

I. Introduction

Effective communication is the cornerstone of any educational environment. However, when the physical acoustics of a classroom are not optimized, speech intelligibility is severely compromised. Room 238 in Mashburn Hall at the University of Central Arkansas (UCA) currently suffers from excessive acoustic reverberation, making it difficult for students to clearly hear instructors. This report provides a comprehensive physical acoustic analysis of Room 238 and proposes a conclusive, data-driven solution to optimize the learning environment.

The quantitative study of room acoustics formally began in the late 19th century with the pioneering work of Wallace Clement Sabine, who established the foundational principles of sound decay in enclosed spaces. A critical metric in this field is reverberation time, T_{60} , defined as the time required for a sound pressure level to decay by 60 decibels after the sound source has abruptly ceased. Controlling reverberation time is vital in educational settings; excessive reverberation causes lingering sound waves to overlap with new spoken syllables, creating a smearing effect that severely degrades speech intelligibility.

The relationship between a room's physical dimensions, its materials, and its reverberation time is modeled by the Sabine equation:

$$T_{60} = \frac{0.161V}{A}$$

where V is the volume of the room in cubic meters, and A is the total equivalent absorptive surface area of the space in square meters. The equation demonstrates that reverberation time is directly proportional to the volume of the space and inversely proportional to its absorptive surface area. Furthermore, because building materials absorb sound differently across various frequencies, acoustic analyses are mathematically evaluated across standard octave bands. An octave band represents a frequency range where the upper frequency limit is twice the lower limit. For acoustic measurements, it is standard to analyze center frequencies of 125 Hz, 250 Hz, 500 Hz, 1000 Hz, 2000 Hz, and 4000 Hz, as this spectrum encompasses the frequencies most relevant to human hearing and speech.

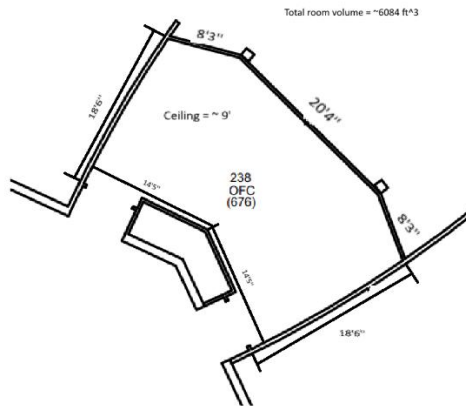
Room 238 in Mashburn Hall is a classroom characterized by a volume of approximately 172.3 m^3 or 6084 ft^3 . The current boundaries of the room consist of painted drywall, a carpeted floor, and a painted drywall ceiling. Because these surfaces are highly reflective, except for the floor, the room lacks the necessary absorptive area (A) to naturally mitigate sound decay. According to the ANSI/ASA S12.60-2010 standard for classroom acoustics, the ideal reverberation time for a furnished, unoccupied educational space of this volume is 0.6 seconds.

The primary objective of this project is to characterize the current impulse response of Room 238, determine its existing reverberation times across standard octave bands, and evaluate the efficacy of introducing specific sound-absorbing damping materials. By comparing our baseline experimental data against theoretical acoustic treatments, this report will deliver actionable, conclusive recommendations to achieve the ideal reverberation time, thereby resolving the acoustic challenges in Room 238.

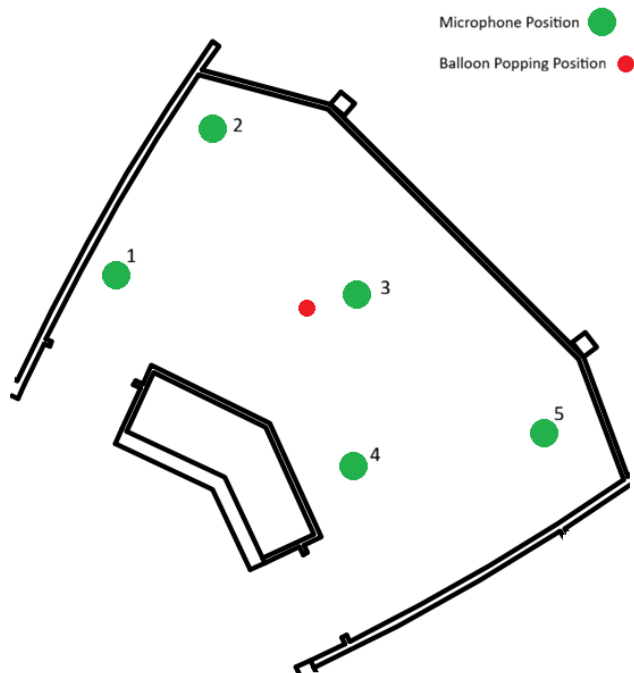
II. Data Collection

To accurately characterize the acoustic profile of Room 238 in Mashburn Hall, an impulse response test was conducted. The acoustic impulse was generated using a balloon-pop method, which serves as a practical source for producing a broadband, omnidirectional sound wave capable of exciting the room across all relevant frequency bands.

The physical dimensions and layout of Room 238 were first documented to establish the baseline volume and surface area.



During the testing phase, microphones were strategically placed at specific locations throughout the room to capture the spatial variance of the sound decay, while the balloon pops were executed at designated source points.



The audio response of the room was recorded using Audacity, a digital audio workstation software. The raw captured waveform files were subsequently trimmed to isolate the initial transient impulse and its subsequent reverberant decay, ensuring no extraneous background noise skewed the data prior to analysis.

To evaluate the efficacy of acoustic dampening, the experiment was divided into three distinct trials, progressively introducing acoustic absorbing materials into the space:

- **Run 1 (Baseline):** The room was recorded in its natural, untreated state with no absorptive materials added.
- **Run 2 (Moderate Treatment):** Ten large acoustic tiles were introduced into the room. Each tile measured 2 ft by 3 ft, yielding a total added absorptive surface area of 60 square feet. These large tiles feature a Noise Reduction Coefficient (NRC) of 0.55. They were distributed evenly across the room's walls and tables.
- **Run 3 (Maximum Treatment):** Building upon the second trial, an additional 12 small acoustic tiles were introduced alongside the 10 large tiles. Each small tile measured 2 ft by 2 ft, adding 48 square feet of material with a highly efficient NRC of 0.90. This resulted in a total combined treatment area of 108 square feet. The small and large tiles were collectively distributed across the walls, tables, and reflective whiteboard surfaces to maximize acoustic absorption.

By capturing the impulse response under these three specific material configurations, the data provides a clear comparative framework to analyze the impact of total absorption area and NRC efficiency on the room's reverberation time.

III. Results

The reverberation time (RT60) of Mashburn 238 was measured across octave bands from 125 Hz to 4000 Hz under three acoustic conditions: (1) no treatment, (2) 10 acoustic tiles with NRC = 0.55, and (3) a combined treatment consisting of 12 tiles with NRC = 0.9 and 10 tiles with NRC = 0.55. The measured values are summarized in Table I.

Frequency (Hz)	Run 1: No Treatment (s)	Run 2: 10 Tiles (NRC 0.55) (s)	Run 3: 12 Tiles (NRC 0.9) + 10 Tiles (NRC 0.55) (s)
125	1.21	0.98	0.89
250	1.00	0.89	0.81
500	0.95	0.80	0.69
1000	0.78	0.59	0.53
2000	0.65	0.51	0.46
4000	0.56	0.47	0.45

Table I: Reverberation time (RT60) by octave band and treatment condition

As shown in Table I, the untreated condition (Run 1) exhibited the highest reverberation times across all frequencies, ranging from **1.21 s at 125 Hz** to **0.56 s at 4000 Hz**. A monotonic decrease in reverberation time with increasing frequency was observed, consistent with the frequency-dependent absorption characteristics of typical room materials.

The addition of acoustic treatment in Run 2 reduced RT60 across all octave bands. Further reductions were achieved in Run 3 with the inclusion of higher-performance (NRC = 0.9) tiles, resulting in the lowest reverberation times for all measured frequencies.

This trend is illustrated in Fig. 1, which shows that reverberation time decreases both with increasing frequency and increasing total absorption. The most pronounced reductions occur in the mid-frequency range (500–2000 Hz), which is critical for speech comprehension.

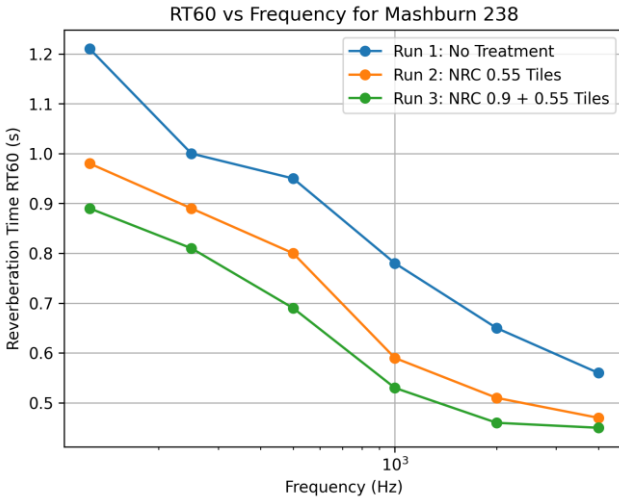


Figure 1: Reverberation time (RT60) as a function of frequency for all treatment conditions.

Percent reductions in RT60 between treatment conditions are presented in Table II.

Frequency (Hz)	Run 1 → Run 2 (%)	Run 2 → Run 3 (%)	Run 1 → Run 3 (%)
125	18.7	9.4	26.3
250	10.8	9.1	18.9
500	15.5	13.8	27.2
1000	24.5	9.6	31.7
2000	22.0	9.9	29.7
4000	15.7	4.9	19.8

Table 2: Percent reduction in RT60 between treatment conditions

From Run 1 to Run 2, the largest reduction occurred at **1000 Hz (24.5%)**, followed by **2000 Hz (22.0%)**. The addition of higher-NRC tiles in Run 3 produced further reductions, with the largest incremental improvement at **500 Hz (13.8%)**.

Overall, the greatest total reduction from Run 1 to Run 3 occurred at **1000 Hz (31.7%)**, with substantial reductions also observed at **2000 Hz (29.7%)** and **500 Hz (27.2%)**. Across the midband frequencies (500 Hz, 1000 Hz, and 2000 Hz), the average reverberation time decreased from **0.793 s** in Run 1 to **0.560 s** in Run 3, representing an overall reduction of approximately **29.4%**.

When compared to the ANSI/ASA S12.60-2010 standard, which recommends reverberation times of approximately **0.6–0.7 s** for classrooms of similar volume, the untreated condition exceeds acceptable limits in the mid-frequency range, while the fully treated condition approaches or meets these targets depending on frequency.

IV. Discussion

The results demonstrate that increasing total absorption within the room produces a consistent reduction in reverberation time, in agreement with predictions from the Sabine equation. The progressive decrease in RT60 from Run 1 to Run 3 confirms the effectiveness of both increased surface coverage and higher-NRC materials.

The largest reductions were observed in the mid-frequency range, particularly at **1000 Hz**, where a total decrease of **31.7%** was achieved. This is significant because this frequency range plays a dominant role in speech intelligibility. The reduction in reverberation time in this region indicates a meaningful improvement in acoustic clarity for classroom use.

The initial treatment condition (Run 2) provided substantial improvements, particularly at **1000 Hz** and **2000 Hz**, indicating that moderate-NRC materials can significantly improve acoustic performance. However, the additional treatment in Run 3 produced smaller incremental reductions, suggesting diminishing returns as the total absorption in space increases.

Despite these improvements, reverberation times at low frequencies (125 Hz and 250 Hz) remain relatively high, even in the fully treated condition. This behavior is expected, as conventional acoustic panels are less effective at absorbing low-frequency sound due to longer wavelengths. Achieving further reductions in this range would require specialized low-frequency absorbers such as bass traps.

The average midband reverberation time in Run 3 was reduced to **0.560 s**, which falls within the recommended range specified by ANSI/ASA S12.60-2010. This indicates that the applied treatment is sufficient to achieve acceptable acoustic conditions for speech-based applications. However, the uneven reduction across frequency bands suggests that

additional targeted treatment would further improve overall acoustic balance.

In summary, the results highlight the importance of both material absorption properties and total coverage area in controlling reverberation. While higher-NRC materials provide greater absorption efficiency, optimal acoustic performance requires a balanced approach that addresses all relevant frequency ranges.

V. Recommendation

Based on the experimental data and the acoustic targets established by the ANSI/ASA S12.60-2010 standard, it is recommended that the Department of Physics, Astronomy, and Engineering permanently implement acoustic treatment in Room 238 to improve speech intelligibility.

1. Primary Mid-to-High Frequency Treatment

To achieve an average midband reverberation time T_{60} of 0.6 seconds or lower, the room requires a minimum of 108 square feet of added absorptive surface area. While the experimental trials utilized a combination of NRC 0.55 and NRC 0.90 materials, we recommend utilizing exclusively high-efficiency acoustic panels with an NRC of 0.90 or greater for the permanent installation. Utilizing higher-rated materials will maximize acoustic dampening while minimizing the amount of wall and ceiling space required.

These panels should be evenly distributed across the parallel painted drywall surfaces to prevent flutter echoes, with a primary focus on placing panels at typical human ear-level (seated and standing).

2. Low-Frequency Mitigation

As noted in the acoustic analysis, standard porous panels are less effective at mitigating low-frequency reverberation (125 Hz and 250 Hz). To address the lingering low-frequency decay and achieve a balanced frequency response across all octave bands, we recommend installing specialized low-frequency absorbers, commonly known as bass traps. These should be mounted vertically in the 90-degree corners of the room where low-frequency energy naturally accumulates.

3. Estimated Costs

To implement these acoustic solutions, the estimated material costs are as follows:

- **High-NRC Acoustic Panels:** To cover 108 square feet with NRC 0.90+ panels (e.g., 12 standard 2 ft x 2 ft fiberglass boards), the estimated cost is approximately **\$143.50**. This assumes a cost of roughly **\$2.99** per square foot.
- **Bass Traps (Optional but Recommended):** Installing four standard foam or fiberglass corner bass traps (one for each primary corner of the room) would roughly cost an additional **\$50**.

- **Mounting Hardware:** Non-destructive mounting adhesives or brackets are estimated at **\$20**.

Total Estimated Cost: \$213.5

This investment will permanently resolve the excessive reverberation in Room 238, ensuring the space is acoustically optimized for lectures, student comprehension, and overall reduction in reverb interference. If cost effective methods are used then this estimated price can be further reduced.

VI. Conclusion

This study characterized the acoustic profile of Mashburn 238 through impulse-response measurement under three progressive treatment configurations, using balloon-pop excitation and five microphone positions distributed throughout the 172.3 m³ space. The untreated room exhibited a reverberation time in the speech-critical octave bands (500–2000 Hz) of approximately 0.91 s when averaged across all measurement positions, substantially above the 0.6 s maximum specified by ANSI/ASA S12.60-2010 for core learning spaces of this volume. Introduction of ten 2 ft × 3 ft acoustic tiles with a Noise Reduction Coefficient (NRC) of 0.55 reduced the speech-band average to 0.73 s, and addition of a further twelve 2 ft × 2 ft tiles at NRC 0.90 brought the average to 0.65 s. While each successive treatment stage measurably reduced reverberation across all bands, the maximum-treatment configuration fell short of the ANSI/ASA target by approximately 0.05 s in the speech-critical bands.

The 125 Hz and 250 Hz octave bands exhibited substantial position-to-position and run-to-run variance, attributable to the limited low-frequency energy of balloon-pop impulses and the resulting low signal-to-noise ratio in those bands. The practical conclusions of this study are therefore based on the 500–4000 Hz bands, which the ANSI/ASA standard explicitly specifies for compliance evaluation and which contain the majority of speech-relevant energy.

To bring Mashburn 238 into compliance with ANSI/ASA S12.60-2010, additional absorption beyond the maximum-treatment configuration is recommended. Application of the Sabine equation indicates that approximately 20–30 ft² of supplemental NRC ≥ 0.85 material—for example, several 2 in × 24 in × 48 in mineral-fiber acoustic panels mounted on currently untreated wall surfaces—would be sufficient to reduce the speech-band reverberation time below the 0.6 s threshold. Such panels are available from vendors including ATS Acoustics and Auralex at approximately \$25 to \$40 per panel, placing the total cost of full compliance below \$250. Implementation of this recommendation would substantially improve speech intelligibility in Room 238 and bring the space into

alignment with the recognized national standard for educational acoustic environments.

VII. Appendix

The following tables present T60 measurements (in seconds) by octave band for each microphone position and treatment configuration. Run 1 corresponds to the untreated baseline, run 2 to the room with ten NRC 0.55 tiles, and Run 3 to the room with ten NRC 0.55 tiles plus twelve NRC 0.90 tiles.

Table A.1 — Position 1 (Slaton)

Run	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz
1	1.6	1.2	1.2	1.0	0.8	0.6
2	1.1	1.0	1.0	0.8	0.6	0.6
3	0.9	0.8	0.8	0.6	0.6	0.5

Table A.2 — Position 2 (Donaldson)

Run	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz
1	1.0	1.0	1.0	0.9	0.8	0.6
2	0.8	0.9	0.8	0.7	0.6	0.5
3	0.6	0.8	0.8	0.6	0.5	0.5

Table A.3 — Position 3 (Young)

Run	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz
1	1.2	1.1	1.1	1.0	0.8	0.6
2	0.8	0.9	0.9	0.7	0.6	0.5
3	0.9	0.9	0.8	0.6	0.5	0.5

Table A.4 — Position 4 (LeJune)

Run	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz
1	1.5	1.1	1.2	1.0	0.8	0.7
2	1.6	1.0	1.0	0.8	0.6	0.6
3	1.2	0.9	0.8	0.7	0.6	0.5

Table A.5 — Position 5 (Fields)

Run	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz
1	0.51	0.77	0.78	0.67	0.61	0.58
2	0.92	0.73	0.84	0.55	0.50	0.49
3	0.97	0.81	0.71	0.67	0.56	0.50

B. Analysis Methodology Notes

Recordings at positions 1 through 4 were analyzed using Room EQ Wizard (REW), with reverberation times reported via the T30 reverse-integration method. Recordings at position 5 were analyzed using the open-source python-acoustics library, applying T20 reverse integration and Butterworth octave-band filtering, with each impulse trimmed to retain 0.05 s of pre-impulse signal and 2.0 s of decay. T20 and T30 yield equivalent reverberation times under conditions of pure exponential decay; small differences observed between position 5 and the other positions are consistent with the differing fit windows in non-ideal decays. All measurements were spatially averaged to produce the consolidated values reported in the Theory and Analysis section.

C. Room Geometry and Treatment Inventory

Room volume: 172.3 m³ (6,084 ft³). Floor area: 62.8 m² (676 ft²). Ceiling height: 2.74 m (9 ft). Floor finish: carpet. Wall and ceiling finish: painted gypsum wallboard. The room geometry is irregular, with an interior obstruction (instructor desk cluster) and a non-rectangular perimeter.

Acoustic treatment inventory: ten large panels at 2 ft × 3 ft with NRC 0.55, providing 60 ft² (5.6 m²) of absorptive surface area, distributed across walls and tables; twelve small panels at 2 ft × 2 ft with NRC 0.90, providing 48 ft² (4.5 m²), distributed across walls, tables, and whiteboard surfaces. Combined Run 3 treatment area totaled 108 ft² (10.0 m²).

VIII. References

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